

Battery State-of-Health Estimation based on Partial Discharge Curves

Ali Irshayyid and Jun Chen, *Senior Member, IEEE*

Abstract—Accurate state-of-health (SOH) estimation is essential for safe operation and reliable fault diagnosis of lithium-ion batteries. However, existing data-driven methods often rely on specific voltage windows or health indicators that may shift with aging, limiting their applicability when partial discharge events are common. This paper proposes a robust SOH estimation framework based on differential capacity (dQ/dV) curves for partial discharge conditions. The framework uses a reference-concatenated input representation that pairs current and initial-cycle dQ/dV curves, enabling the model to learn degradation-related electrochemical changes. To address incomplete discharge data, a history-based imputation strategy reconstructs missing curve segments using earlier cycles. An attention-based feedforward neural network (FNN-Attention) is developed to capture complex dependencies across voltage levels in the dQ/dV curves. Numerical studies are performed using the MIT-Stanford battery aging dataset, where the proposed FNN-Attention achieves 0.38% root mean square percentage error (RMSPE), an 11.6% improvement over the state-of-the-art reported in literature. With partial discharge curves retaining only 30% (70% of the dQ/dV curve is missing) of the full curve, the FNN-Attention maintains 0.47% RMSPE using history imputation and 0.89% without requiring historical data. The results demonstrate the effectiveness and practical applicability of the proposed framework for real-world battery management systems.

I. INTRODUCTION

Lithium-ion batteries have become the dominant energy storage solution for electric vehicles (EVs) and grid-connected energy storage systems due to their favorable characteristics [1], [2]. However, battery performance inevitably degrades through complex aging process that affects both capacity and internal resistance over time [3]–[5]. Accurate monitoring of battery state-of-health (SOH) is therefore crucial for ensuring safe operation, accurate state-of-charge (SOC) estimation, and reliable fault diagnosis [6]–[8]. The battery SOH is typically defined as the ratio of the current capacity to the initial rated capacity, with a battery generally considered to have reached end-of-life when its capacity drops below 80% of its nominal value [9].

Generally, SOH estimation methods can be classified into experimental calibration, model-based state estimation, and data-driven methods [10]. Experimental calibration methods, such as impedance spectroscopy [11] and full cycling [12], provide accurate measurements but require specialized equipment and extended testing periods. Model-based approaches utilize mathematical representations of battery

behavior paired with recursive filtering algorithms (e.g., H-infinity filters, particle filters, extended Kalman filters) to track capacity fade which offers interpretability but depends heavily on model accuracy, which restricts online applicability [13].

Data-driven methods have emerged as a promising alternative for SOH estimation due to their ability to capture complex nonlinear relationships without requiring explicit physical models [6], [14], [15]. Various machine learning techniques have been applied, including recurrent architectures (RNN, LSTM) [16], convolutional neural networks [17], Gaussian process regression [18], and ensemble learning methods [19].

An important choice of the data-driven SOH estimation is the selection of appropriate health indicators (HIs) that correlate well with battery degradation. Direct measurement features such as charging time, voltage curves, and temperature profiles have been widely used in literature. Additionally, differential analysis methods, particularly incremental capacity (IC) and differential voltage (DV) analysis, have proven effective for extracting degradation-sensitive features from battery cycling data [20]. The IC/DV curves reveal electrochemical phase transitions and their evolution with aging provides rich information about the underlying degradation process [21]. Various voltage-window-based HIs have been proposed, such as the Voltage Variance during Equal Time Interval (VETI) [22], the Degradation Speed Ratio (DSR) [23], and the Duration of Same Charging Voltage Range (DSCVR) [19]. While effective, these share a key limitation: they require the battery to traverse a specific voltage window during charging to extract the HI. When the charging process does not cover that window, the HI becomes unreliable, limiting applicability in real-world settings where partial and random charging are common.

To address random charging voltage ranges, [24] employed IC analysis with a rule-based three-case strategy to handle partial charging voltage range. However, its reliance on detecting specific IC regions limits generalizability across cells and aging. Furthermore, [25] proposed a single SOC-dependent HI named Resistance-Time Constant (RTC) for data-driven SOH estimation, which achieved 0.43% estimation percentage error. However, the approach poses several limitations. First, it is contingent on the accuracy of the SOC estimation to determine when to start and stop the time measurement, leading to potential error accumulation over time. Second, it relies on a single HI that may migrate or shift with aging. Generally, methods that depend on specific operating regions to maximize estimation accuracy inherently constrains the generalizability of these methods, as certain

This work is supported in part by National Science Foundation through Award #2430374. *J. Chen is the corresponding author.*

Ali Irshayyid and Jun Chen are with the Department of Electrical and Computer Engineering, Oakland University, Rochester, MI 48309 USA (e-mail: {aliirshayyid, junchen}@oakland.edu).

HIs may diminish, migrate, or disappear entirely as the battery degrades (e.g., the IC peak used in [24] disappears after extended cycling).

To address the aforementioned limitations, this paper proposes a dQ/dV-based SOH estimation framework for partial discharge conditions. The main contributions are:

- We propose a reference-concatenated input representation that combines current and initial-cycle dQ/dV curves to capture degradation-related electrochemical changes.
- We propose a practical history-based imputation strategy that reconstructs missing dQ/dV segments using recent cycles, avoiding predetermined full discharge requirements.
- We design an FNN-Attention model that captures voltage-level dependencies and supports variable-length partial curves. On the MIT-Stanford/Toyota battery aging dataset, the proposed model achieves 0.38% RM-SPE on full curves, improves the best reported baseline by 11.6%, and maintains 0.89% RMSPE with 70% missing data without historical imputation.

The remainder of this paper is organized as follows. Section II presents the data preparation stage while the curve construction stage in addition to the history-based imputation strategy are detailed in Section III. Section IV details the proposed and baseline model architectures investigated for SOH estimation. Section V outlines the experimental setup and discusses the experimental results. Finally, Section VI concludes the paper and presents future research directions.

II. DATA PREPARATION

This section describes the data preparation stage of the proposed SOH estimation framework. We present the battery dataset, the preprocessing steps, and the feature extraction process.

A. Battery Dataset

This study uses the MIT-Stanford/Toyota battery aging dataset, focusing on the same 17 A123 APR18650M1A LFP cells used in [25] for fair comparison. The cells were aged at 30 °C under fast-charging protocols, charged through CC-CV operation to 3.6 V with a C/50 cutoff, and discharged at 4C to 2.0 V. The selected cells cover diverse degradation patterns, with cycle lives from 635 to 1226 cycles before reaching 80% capacity.

B. Dataset Processing

For each discharge cycle, dQ/dV curves are computed from voltage-capacity measurements after removing invalid samples and averaging duplicated voltage values (see [25] for similar treatment). Each curve is sampled at 131 voltage levels from 2.2 V to 3.5 V with 10 mV resolution, producing the fixed-dimensional inputs used for SOH estimation.

III. CURVE CONSTRUCTION

This section describes the differential curve construction stage of the proposed SOH estimation framework. We present the full curve representation and the proposed *history-based imputation strategy* for handling partial discharge scenarios.

To enable the model to learn degradation-sensitive features, each cycle's dQ/dV curve is concatenated with the reference curve from the first available cycle. This pairing allows the model to detect relative changes, such as peak shifts, amplitude reductions, and shape distortions that correlate with capacity fade. The feature vector for each cycle is defined as.

$$\mathbf{x} = \left[\frac{dQ}{dV} \Big|_{V_1}^{(1)}, \dots, \frac{dQ}{dV} \Big|_{V_L}^{(1)}, \frac{dQ}{dV} \Big|_{V_1}^{(k)}, \dots, \frac{dQ}{dV} \Big|_{V_L}^{(k)} \right]^T \in \mathbb{R}^{262}, \quad (1)$$

where (k) and (1) represent the current and the first (reference) cycle, respectively, and $V_i \in [2.2V \ 2.21V \ \dots \ 3.5V]$. The SOH is calculated as the ratio of the measured discharge capacity to the nominal capacity C_0 as follows:

$$SOH_k = \frac{C_k}{C_0}, \quad (2)$$

where SOH_k and C_k are the state-of-health and capacity at cycle k , respectively. The final dataset consists of 15,736 samples, each containing a 262-dimensional feature vector paired with its corresponding SOH value. Because complete discharge cycles are rarely available in practical BMS operation, partial dQ/dV curves may miss degradation-sensitive regions. We address this using history-based imputation, which fills missing voltage-level values with the most recent available measurements from previous cycles. This is justified by the gradual evolution of battery degradation, which makes recent dQ/dV measurements a reasonable approximation of the current cycle.

Formally, for a partial discharge curve at cycle k where dQ/dV values are unavailable at certain voltage levels $V_{\text{missing}} \subset \{V_1, \dots, V_L\}$, we impute the missing values using corresponding measurements from cycle $k - \Delta$:

$$\frac{dQ}{dV} \Big|_{V_i}^{(k)} \leftarrow \frac{dQ}{dV} \Big|_{V_i}^{(k-\Delta)}, \quad \forall V_i \in V_{\text{missing}}, \quad (3)$$

where Δ denotes the temporal offset such that the most recent V_i is available in $k - \Delta$. This formulation preserves the available current-cycle information while imputing gaps with the most historically-reliable approximation. The effectiveness of this imputation strategy depends on factors such as the amount of missing data within the current cycle and the age of the available historical data used for imputation (i.e., how many cycles separate the current and offset cycles). A systematic evaluation of these factors is presented in Section V-A.

IV. NEURAL NETWORK ARCHITECTURES FOR SOH ESTIMATION

We use two data-driven deep learning architectures to estimate SOH from the processed features dQ/dV : a feed-forward neural network (FNN) and an attention-based FNN.

A. Feedforward Neural Network

The baseline FNN uses a 262-dimensional input formed by concatenating the reference and current dQ/dV curves. It consists of four hidden layers with dimensions [128, 128, 64, 32], followed by a scalar SOH output. GELU activation functions, [26], are used in the first three hidden layers and SiLU, [27], in the final hidden layer.

B. FNN-Attention

The proposed FNN-Attention architecture, originally proposed in our previous work for vehicle platooning control [28], extends the FNN model by incorporating a multi-head self-attention mechanism to capture complex dependencies across voltage levels in the dQ/dV curves. The architecture processes the differential capacity data as a sequence of voltage-indexed inputs. For each voltage level v_i where $i \in \{1, 2, \dots, L\}$, the corresponding dQ/dV values from both the reference and current cycles are processed through a shared feature extractor layer. The extracted features are then concatenated with their respective voltage levels:

$$\mathbf{h}_i = \left[f_{\theta} \left(\left. \frac{dQ}{dV} \right|_{v_i}^{(1)}, \left. \frac{dQ}{dV} \right|_{v_i}^{(k)} \right), v_i \right]^T, \quad (4)$$

where $f_{\theta}(\cdot)$ denotes the feature extractor. The concatenation of voltage-level values serves as an implicit positional encoding, providing the attention layer with the sequence of the discharge curve data. This step will be critically important when working with partial curves.

The resulting sequence of feature vectors $\{\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_L\}$ is passed through a layer normalization module before entering the multi-head self-attention block. The multi-head attention mechanism computes:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^{out}, \quad (5)$$

where each attention head is computed as:

$$\text{head}_j = \text{softmax} \left(\frac{Q_j K_j^T}{\sqrt{d_k}} \right) V_j, \quad (6)$$

where, Q_j , K_j , and V_j are the query, key, and value projections for the j -th head, d_k is the dimension of the key vectors, and W^{out} is the output projection matrix. The multi-head formulation enables the model to have each head focusing on a small subspace and these insights are combined using W^{out} . A residual connection is employed around the attention block to preserve the original feature information and to prevent the vanishing gradient problem [29].

V. NUMERICAL RESULTS

This section presents the experimental results evaluating the proposed FNN-Attention SOH estimation framework under both complete and partial curves scenarios.

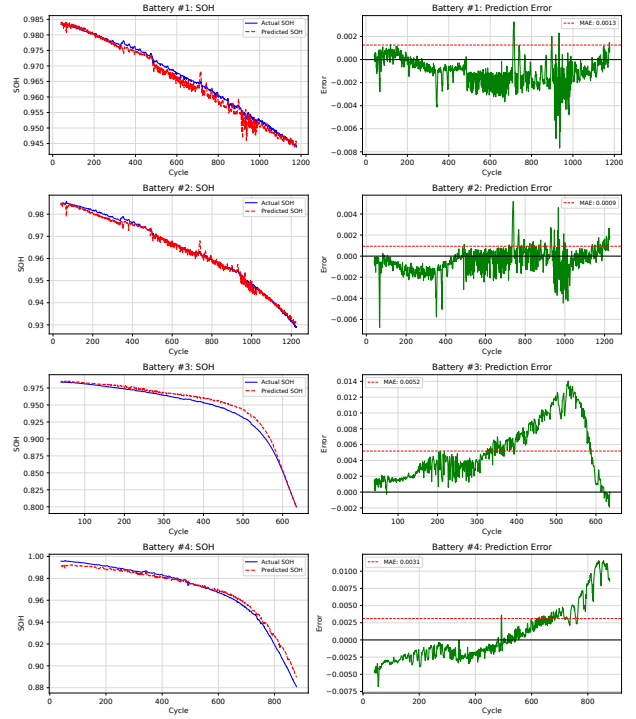


Fig. 1. Performance evaluation of the proposed model on test batteries #1, 2, 3, and 4. *Left*: True vs. predicted SOH. *Right*: Prediction error.

A. Experimental Setup

Experiments were conducted using the battery aging dataset and cell selection described in Section II-A to ensure generalization to unseen batteries. All models were trained exclusively on complete curves and tested on both complete and partial curves as described in Section III. Experiments were conducted on a standard desktop computer with an NVIDIA RTX 3070 Ti GPU.

Partial discharge scenarios are simulated by randomly selecting 70% of cycles, excluding the first cycle of each battery, and removing randomly located contiguous dQ/dV segments. Four maximum missing-data levels are considered: 20%, 30%, 50%, and 70%. For history imputation, missing values are replaced using measurements from 1, 3, or 6 cycles earlier, resulting in twelve partial-curve scenarios. The root mean square percentage error (RMSPE) and mean absolute error (MAE) are selected as evaluation metrics as follows:

$$\text{RMSPE} = \sqrt{\frac{1}{N} \sum_{k=1}^N \left(\frac{\text{SOH}_k - \hat{\text{SOH}}_k}{\text{SOH}_k} \right)^2} \times 100\% \quad (7)$$

$$\text{MAE} = \frac{1}{N} \sum_{k=1}^N |\text{SOH}_k - \hat{\text{SOH}}_k|, \quad (8)$$

where SOH_k is the true value, $\hat{\text{SOH}}_k$ is the prediction value, and N is the total number of cycles of the testing cell.

TABLE I
ESTIMATION ACCURACY UNDER DIFFERENT PARTIAL-CURVE SCENARIOS FOR THE ADOPTED MODELS.

Missing Curve Data	Full curve		1-cycle Offset		3-cycle Offset		6-cycle Offset	
	FNN	FNN-Att.	FNN	FNN-Att.	FNN	FNN-Att.	FNN	FNN-Att.
<i>RMSPE (%)</i>								
0% (Full curve)	0.39	0.38						
20% (30 pts)	—		0.41	0.39	0.41	0.39	0.43	0.41
30% (40 pts)	—		0.41	0.39	0.42	0.40	0.45	0.41
50% (66 pts)	—		0.45	0.40	0.50	0.41	0.59	0.44
70% (91 pts)	—		0.56	0.47	0.68	0.54	0.93	0.67
<i>MAE</i>								
0% (Full curve)	0.0027	0.0026						
20% (30 pts)	—		0.0028	0.0026	0.0028	0.0027	0.0030	0.0028
30% (40 pts)	—		0.0029	0.0026	0.00297	0.0026	0.0031	0.0027
50% (66 pts)	—		0.0030	0.0026	0.0034	0.0027	0.0041	0.0029
70% (91 pts)	—		0.0035	0.0029	0.0046	0.0033	0.0062	0.0039

TABLE II
COMPUTATIONAL COMPLEXITY COMPARISON OF THE PROPOSED MODELS

Model	Parameters	Inference Time (ms)	Training Time (s)	Variable-Length Input
FNN	77,057	0.161	~25	No
FNN-Attention	4,521	1.99	~5400	Yes

TABLE III
THE PROPOSED FNN-ATTENTION MODEL PERFORMANCE ON TEST CELLS.

Battery No.	Battery Label	RMSPE (%)	MAE
1	'3.6C(80%)-3.6C'	0.16	0.0013
2	'4C(80%)-4C'	0.13	0.0009
3	'4.8C(80%)-4.8C'	0.68	0.0052
4	'5.4C(40%)-3.6C'	0.44	0.0031
5	'5.4C(50%)-3C'	0.32	0.0021
6	'5.4C(50%)-3.6C'	0.46	0.0039
7	'6C(30%)-3.6C'	0.53	0.0040
8	'6C(40%)-3.6C'	0.21	0.0016

B. Full dQ/dV Curve Performance

This section evaluates the SOH estimation accuracy of the proposed FNN-Attention model and the baseline FNN when complete dQ/dV curves are available.

Table I (first column) summarizes the performance metrics for both models on full discharge curves. The FNN-attention achieves an RMSPE of 0.38% and MAE of 0.0026, outperforming the baseline FNN (RMSPE: 0.39%, MAE: 0.0027). Table II summarizes the computational complexity of the used models. The FNN model achieves the fastest inference time (0.161 ms) and shortest training time (~25 s), making it well-suited for deployment on resource-constrained embedded systems. The FNN-Attention model has the fewest trainable parameters (4,521) and is the only network that supports variable-length inputs, enabling SOH estimation from partial discharge curves even without the need for historical data imputation. However, this flexibility comes at the cost of longer training time (~90 min) due to the sequential processing of voltage-indexed inputs.

TABLE IV
PERFORMANCE COMPARISON WITH BASELINE METHODS (AS REPORTED IN [25]).

Method	RMSPE (%)	MAE
FNN-Attention (Proposed)	0.38	0.0026
RTC [25]	0.43	0.0027
VVETI [22]	3.63	0.0238
DSR [23]	3.17	0.0226
ICE [24]	3.76	0.0259
TVR [19]	1.20	0.0094

As shown in Table III and Figures. 1 and 2, FNN-Attention maintains consistent accuracy across all unseen test cells, with RMSPE below 0.68% for every battery. Compared with existing methods in Table IV, the proposed model achieves the lowest RMSPE of 0.38%, improving over RTC [25] by 11.6% and substantially outperforming VVETI, DSR, ICE, and TVR.

C. Partial dQ/dV Curve Performance with History Imputation Strategy

In practical battery management applications, complete discharge cycles are rarely available due to user behavior and operational constraints. This section evaluates the robustness of the used models under partial discharge scenarios where missing dQ/dV segments are imputed using historical cycle data. The assumption is that battery degradation occurs gradually, and therefore, dQ/dV characteristics from recent cycles will provide reasonable approximations for missing segments. Table I presents comprehensive results across twelve experimental scenarios, combining four maximum removal thresholds (20%, 30%, 50%, and 70%) with three

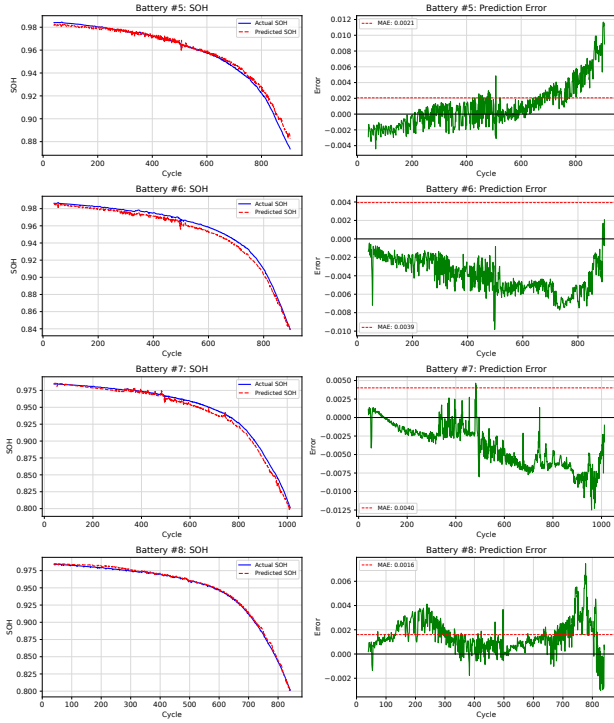


Fig. 2. Performance evaluation of the proposed model on test batteries #5, 6, 7, and 8. *Left*: True vs. predicted SOH. *Right*: Prediction error.

temporal offsets (1, 3, and 6) for adopted models. All models were trained exclusively on complete curves and evaluated on partial curves with missing segments being replaced by an older cycle's values. The results demonstrate that both models maintain robust performance when historical data is available to impute missing segments. With 1-cycle offset and 20% missing data, the proposed FNN-Attention achieves an RMSPE of 0.39%, representing only a 2.6% relative increase from the full-curve baseline of 0.38%. The baseline FNN shows a similar trend, with RMSPE increasing from 0.39% to 0.41%. This resilience is expected since the 1-cycle offset provides the most recent historical information, which closely approximates the current cycle's characteristics due to the gradual nature of battery aging. As the severity of data loss increases, performance degradation becomes more pronounced but remains within acceptable bounds. At 50% missing data with 1-cycle offset, FNN-Attention achieves 0.40% RMSPE, while FNN achieves 0.45%. Even under the most challenging scenario of 70% missing data, both models produce usable estimates (FNN-Attention at 0.47% RMSPE and FNN at 0.56%). These results indicate that the models can effectively leverage the available partial information combined with historical imputation to maintain estimation accuracy.

The temporal offset analysis reveals the impact of historical data freshness on the SOH estimation performance. FNN-Attention's RMSPE increases from 0.47% (1-cycle) to 0.54% (3-cycle) to 0.67% (6-cycle). The baseline FNN exhibits greater sensitivity to temporal offset, with RMSPE increasing from 0.56% to 0.68% to 0.93% across the same

conditions. The proposed FNN-Attention model consistently outperforms the baseline FNN across all partial-curve scenarios, demonstrating superior robustness to incomplete data. Notably, under severe data loss conditions (70% missing data with 6-cycle offset), FNN-Attention achieves 0.67% RMSPE compared to FNN's 0.93% which is a 27.95% relative improvement. This enhanced robustness can be attributed to the attention mechanism's ability to dynamically weight the importance of available dQ/dV points and capture long-range dependencies across the voltage spectrum. Collectively, the results validate the proposed history-imputation strategy as an effective approach for handling incomplete discharge data in practical deployments.

D. Partial dQ/dV Curve Performance without History Imputation

In the previous section, we demonstrated effective SOH estimation when historical cycle data is available to impute missing dQ/dV segments. However, practical deployment scenarios may arise where historical data is unavailable, such as during initial battery deployment, or when batteries are transferred between different management systems. This section investigates the performance under such challenging scenarios (without history availability), focusing on two approaches for handling such cases. The first approach utilizes the FNN-Attention model, which processes only the available dQ/dV points along with their corresponding voltage levels. This evaluates the attention mechanism's ability to handle variable-length sequences without reconstruction or padding. The attention mechanism's inherent flexibility allows it to operate on inputs of arbitrary length, while the explicit voltage concatenation provides positional context that enables meaningful feature extraction from incomplete curves. The second approach employs the baseline FNN model with linear extrapolation, where missing dQ/dV segment is filled by linear extrapolation using the closest 10 available points. This serves as a benchmark representing conventional approaches that rely on data reconstruction to satisfy fixed-length input requirements.

Table V evaluates cases where historical data are unavailable. FNN-Attention directly processes the available voltage-indexed dQ/dV points and maintains 0.89% RMSPE even with 70% missing data. In contrast, FNN with linear extrapolation degrades sharply, reaching 33.6% RMSPE under the same condition. This demonstrates the advantage of variable-length attention-based processing over fixed-input reconstruction.

Comparing FNN-Attention's performance without history (Table V) to its performance with history-imputation (Table I) reveals the value of historical data when available. At 70% missing data, FNN-Attention achieves 0.47% RMSPE with 1-cycle history-imputation versus 0.89% without history-imputation, i.e., a 47% relative improvement. This comparison suggests that while the attention-based architecture provides substantial robustness to missing data on its own, incorporating historical information when available further enhances estimation accuracy.

TABLE V

SOH ESTIMATION UNDER PARTIAL DISCHARGE CURVES (NO HISTORY REPLACEMENT).

Missing Curve Data	FNN-Att.		FNN + Lin. Extrap.	
	MAE	RMSPE (%)	MAE	RMSPE (%)
0% (0 pts)	0.0026	0.38	0.0027	0.39
20% (30 pts)	0.0027	0.40	0.0094	1.10
30% (40 pts)	0.0028	0.42	0.0280	3.12
50% (66 pts)	0.0034	0.50	0.9800	10.3
70% (91 pts)	0.0054	0.89	3.2100	33.6

VI. CONCLUSION

This paper proposes a dQ/dV-based SOH estimation framework for partial discharge conditions. By combining reference-concatenated inputs, history-based imputation, and an FNN-Attention architecture, the method captures degradation-related voltage-level dependencies while remaining robust to missing curve segments. Experimental results on the MIT-Stanford/Toyota battery aging dataset demonstrate that FNN-Attention achieves 0.38% RMSPE on complete curves, representing an 11.6% improvement over the best existing baseline approach. Under partial discharge scenarios, the model achieves 0.47% RMSPE with 70% missing data when 1-cycle history imputation is available, and 0.89% RMSPE under the same missing-data level without historical data. These results demonstrate the framework's potential for real-world BMS applications where complete discharge cycles are rarely available.

REFERENCES

- [1] X. Han, M. Ouyang, L. Lu, J. Li, Y. Zheng, and Z. Li, "A comparative study of commercial lithium ion battery cycle life in electrical vehicle: Aging mechanism identification," *Journal of power sources*, vol. 251, pp. 38–54, 2014.
- [2] J. Chen, A. Behal, Z. Li, and C. Li, "Active battery cell balancing by real time model predictive control for extending electric vehicle driving range," *IEEE Trans. Auto. Sci. Eng.*, vol. 21, no. 3, pp. 4003–4015, July 2024.
- [3] D.-I. Stroe, M. Swierczynski, A.-I. Stroe, R. Laerke, P. C. Kjaer, and R. Teodorescu, "Degradation behavior of lithium-ion batteries based on lifetime models and field measured frequency regulation mission profile," *IEEE Transactions on Industry Applications*, vol. 52, no. 6, pp. 5009–5018, 2016.
- [4] H. Tian, P. Qin, K. Li, and Z. Zhao, "A review of the state of health for lithium-ion batteries: Research status and suggestions," *Journal of Cleaner Production*, vol. 261, p. 120813, 2020.
- [5] J. Chen, Z. Zhou, Z. Zhou, X. Wang, and B. Liaw, "Impact of battery cell imbalance on electric vehicle range," *Green Energy and Intelligent Transportation*, vol. 1, no. 3, pp. 1–8, December 2022.
- [6] X. Hu, Y. Che, X. Lin, and S. Onori, "Battery health prediction using fusion-based feature selection and machine learning," *IEEE Transactions on Transportation Electrification*, vol. 7, no. 2, pp. 382–398, 2020.
- [7] G. L. Plett, *Battery management systems, Volume II: Equivalent-circuit methods*. Artech House, 2015.
- [8] Y. Wang, J. Tian, Z. Sun, L. Wang, R. Xu, M. Li, and Z. Chen, "A comprehensive review of battery modeling and state estimation approaches for advanced battery management systems," *Renewable and Sustainable Energy Reviews*, vol. 131, p. 110015, 2020.
- [9] S. Yang, C. Zhang, J. Jiang, W. Zhang, L. Zhang, and Y. Wang, "Review on state-of-health of lithium-ion batteries: Characterizations, estimations and applications," *Journal of Cleaner Production*, vol. 314, p. 128015, 2021.
- [10] X. Sui, S. He, S. B. Vilsen, J. Meng, R. Teodorescu, and D.-I. Stroe, "A review of non-probabilistic machine learning-based state of health estimation techniques for lithium-ion battery," *Applied Energy*, vol. 300, p. 117346, 2021.
- [11] Y. Cui, P. Zuo, C. Du, Y. Gao, J. Yang, X. Cheng, Y. Ma, and G. Yin, "State of health diagnosis model for lithium ion batteries based on real-time impedance and open circuit voltage parameters identification method," *Energy*, vol. 144, pp. 647–656, 2018.
- [12] Y. Tan and G. Zhao, "Transfer learning with long short-term memory network for state-of-health prediction of lithium-ion batteries," *IEEE Transactions on Industrial Electronics*, vol. 67, no. 10, pp. 8723–8731, 2019.
- [13] C. Chen, R. Xiong, and W. Shen, "A lithium-ion battery-in-the-loop approach to test and validate multiscale dual h infinity filters for state-of-charge and capacity estimation," *IEEE Transactions on power Electronics*, vol. 33, no. 1, pp. 332–342, 2017.
- [14] A. Irshayyid and J. Chen, "GNN-based surrogate model for reconfigurable battery packs," in *IEEE Conference on Control Technology and Applications*, San Diego, CA, August 25–27, 2025.
- [15] —, "Surrogate model for reconfigurable battery packs using graph neural networks," *IEEE Transactions on Industrial Informatics*, 2026.
- [16] Y. Zhang, R. Xiong, H. He, and M. G. Pecht, "Long short-term memory recurrent neural network for remaining useful life prediction of lithium-ion batteries," *IEEE Transactions on Vehicular Technology*, vol. 67, no. 7, pp. 5695–5705, 2018.
- [17] J. Chen, M. Manivanan, J. Duque, P. Kollmeyer, S. Panchal, O. Gross, and A. Emadi, "A convolutional neural network for estimation of lithium-ion battery state-of-health during constant current operation," in *2023 IEEE transportation electrification conference & expo (ITEC)*. IEEE, 2023, pp. 1–6.
- [18] X. Li, C. Yuan, X. Li, and Z. Wang, "State of health estimation for li-ion battery using incremental capacity analysis and gaussian process regression," *Energy*, vol. 190, p. 116467, 2020.
- [19] B. Gou, Y. Xu, and X. Feng, "An ensemble learning-based data-driven method for online state-of-health estimation of lithium-ion batteries," *IEEE Transactions on Transportation Electrification*, vol. 7, no. 2, pp. 422–436, 2020.
- [20] D. Anseán, V. M. García, M. González, C. Blanco-Viejo, J. C. Viera, Y. F. Pulido, and L. Sánchez, "Lithium-ion battery degradation indicators via incremental capacity analysis," *IEEE Transactions on Industry Applications*, vol. 55, no. 3, pp. 2992–3002, 2019.
- [21] M. Berecibar, M. Garmendia, I. Gandiaga, J. Crego, and I. Villarreal, "State of health estimation algorithm of lifepo4 battery packs based on differential voltage curves for battery management system application," *Energy*, vol. 103, pp. 784–796, 2016.
- [22] W. Liu, Y. Xu, and X. Feng, "A hierarchical and flexible data-driven method for online state-of-health estimation of li-ion battery," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 12, pp. 14 739–14 748, 2020.
- [23] I. Alvarez-Monteserin and M. Á. Sanz-Bobi, "An online fade capacity estimation of lithium-ion battery using a new health indicator based only on a short period of the charging voltage profile," *IEEE Access*, vol. 10, pp. 11 138–11 146, 2022.
- [24] C. Zhang, Y. Kang, B. Duan, Z. Zhou, Q. Zhang, Y. Shang, and A. Chen, "An adaptive battery capacity estimation method suitable for random charging voltage range in electric vehicles," *IEEE Transactions on Industrial Electronics*, vol. 69, no. 9, pp. 9121–9132, 2021.
- [25] E. Kheirkhah-Rad, A. Parvareh, M. Moeini-Aghtaie, and P. Dehghanian, "A data-driven state-of-health estimation model for lithium-ion batteries using referenced-based charging time," *IEEE Transactions on Power Delivery*, vol. 38, no. 5, pp. 3406–3416, 2023.
- [26] D. Hendrycks, "Gaussian error linear units (gelus)," *arXiv preprint arXiv:1606.08415*, 2016.
- [27] S. Elfving, E. Uchibe, and K. Doya, "Sigmoid-weighted linear units for neural network function approximation in reinforcement learning," *Neural networks*, vol. 107, pp. 3–11, 2018.
- [28] A. Irshayyid and J. Chen, "Highway merging control using multi-agent reinforcement learning," in *IEEE International Conference on Computing and Machine Intelligence*, Mount Pleasant, MI, April 13–14, 2024.
- [29] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.